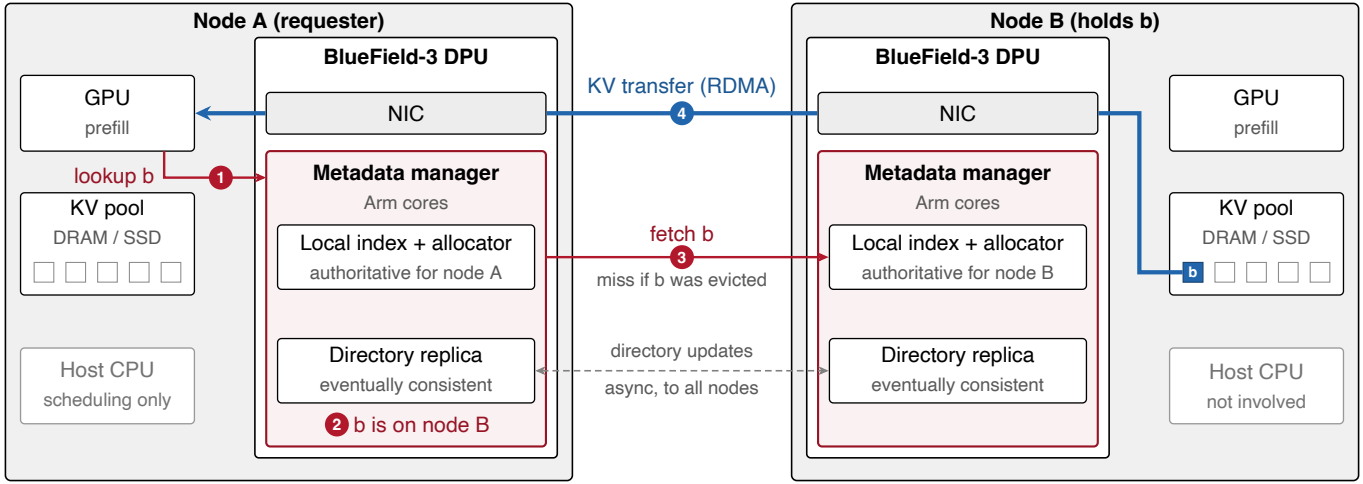
Node A (requester)
BlueField-3 DPU
GPU
prefill
NIC
KV transfer (RDMA)
4
Node B (holds b)
BlueField-3 DPU
NIC
GPU
prefill
lookup b
1
Metadata manager
Arm cores
Local index + allocator
authoritative for node A
fetch b
3
miss if b was evicted
Metadata manager
Arm cores
Local index + allocator
authoritative for node B
KV pool
DRAM / SSD
KV pool
DRAM / SSD
b
Directory replica
eventually consistent
directory updates
async, to all nodes
Directory replica
eventually consistent
Host CPU
scheduling only
2 b is on node B
Host CPU
not involved